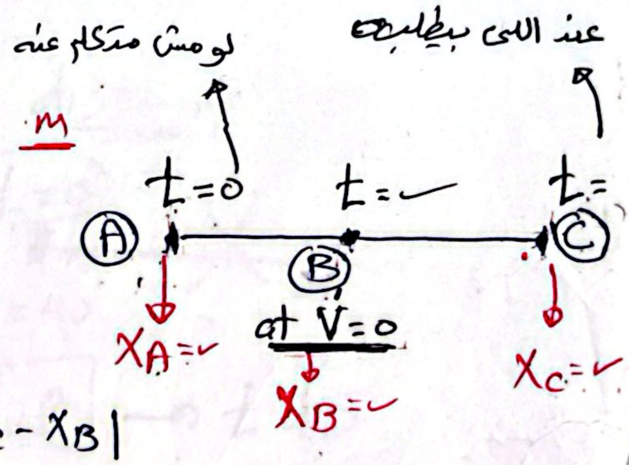
Straight line motion

- Position (x) or (S) : m (origin) موضع الجسم من نقطة
- Velocity (V.) : m/sec
- acceleration (a.) : m/sec²
- displacement (ΔX) or (ΔS) : m

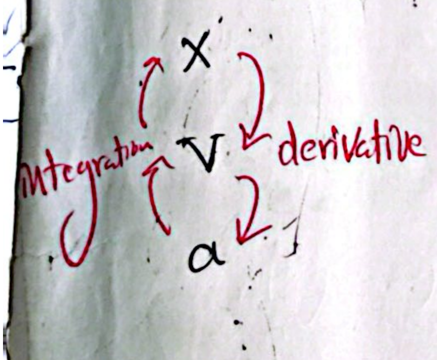
$$\Delta X = X_C - X_A$$

- distance (d) : m

$$d = |X_B - X_A| + |X_C - X_B|$$



(معادلات في time)



معادلات في (x) أو (V)

$$\int a\,dx = \int V\,dV \longrightarrow (X, V)$$

$$\int a\,dt = \int dV \longrightarrow (t, V)$$

$$\int V\,dt = \int dx \longrightarrow (t, X)$$

Body starts from rest → $(V_0 = 0)$ at $t = 0$ ← initial

at origin → $(X = 0)$

The position of a particle which moves along a straight line is defined by the relation, $x = t^3 - 6t^2 - 15t + 40$, where x is expressed in meter and t in seconds. Determine:

(a) the time at which the velocity will be zero
(b) the position and distance traveled by the particle at that time and displacement
(c) the acceleration of the particle at that time
(d) the distance and displacement traveled by the particle from $t = 4s$ to $t = 6s$
(e) the net displacement traveled by the particle when $a = 0$

$$X = t^3 - 6t^2 - 15t + 40$$

$$V = 3t^2 - 12t - 15$$

$$a = 6t - 12$$

(a) $t = ?? \rightarrow V = 0$

$$0 = 3t^2 - 12t - 15$$

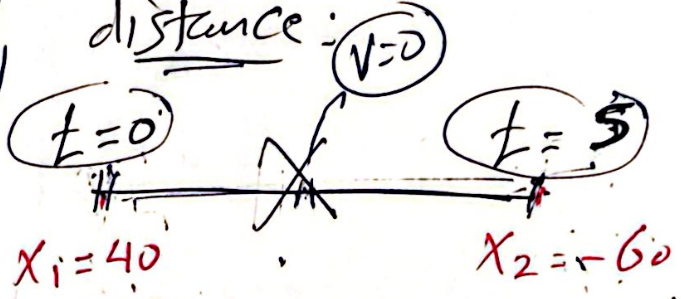
$\boxed{t = 5}$ $\quad \cancel{t = -1}$

(b) $X = ??$ } $\rightarrow t = 5$
distance = ??

Position:

$X = -60\,m$

distance:



$\boxed{V = 0} \rightarrow t = 5$

$t = 0 \rightarrow X_1 = 40$

$t = 5 \rightarrow X_2 = -60$

distance $= |X_2 - X_1|$

$= |-60 - 40| = \boxed{100\,m}$

displacement

$= -60 - 40 = \boxed{-100\,m}$

(c) $a = ?? \rightarrow t = 5$

$a = 6(5) - 12 = 18\,m/s^2$

(d) $t=4$ $\quad t=5$ $\quad t=6$

$\boxed{V=0}$ (at $t=5$)

$X_1=-52$ $\quad X_2=-60$ $\quad X_3=-50$

distance $= |X_3-X_2| + |X_2-X_1|$.

$= |-50+60| + |-60+52| = 10+8$

$= \underline{18m}$

displacement $= X_3-X_1 = -50+52$

$= \underline{2m}$

(e) displacement $= ??$ when $\boxed{a=0}$

$0 = 6t-12$

$\boxed{t = 2sec}$

$t=0$ $\quad t=2$

$X_1=40$ $\quad X_2=-6$

disp $= -6-40$

$= \underline{-46m}$

$\int 0 = (+c)$ $\quad x \to s,\ v,\ a$ $\quad st \to (5)$

The acceleration of a particle as it moves along a straight line is given by $a = (2t - 1)\ \text{m/s}^2$, where t is in seconds. If s = 1 m and v = 2 m/s when t = 0, Determine the particle's velocity and position when t = 6 s. Also, determine the total distance the particle travels during this time period.

$$\boxed{a = 2t - 1}$$

$$\boxed{x = 1 \to V = 2 \to t = 0}$$

$$V\Big|_2^V = \frac{2t^2}{2} - t\Big|_0^t$$

$$V - 2 = t^2 - t - (0)$$

$$\boxed{V = t^2 - t + 2}$$

$$x\Big|_1^x = \frac{t^3}{3} - \frac{t^2}{2} + 2t\Big|_0^t$$

$$x - 1 = \frac{t^3}{3} - \frac{t^2}{2} + 2t - 0$$

$$\boxed{X = \frac{t^3}{3} - \frac{t^2}{2} + 2t + 1}$$

(a) $V = ??\ ,\ X = ??$

$t = 6\ \text{sec}$

$V = 32\ \text{m/sec}$

$X = 67\ \text{m}$

(b) distance

$t = 0$ $\quad X \quad$ $t = 6$

$\boxed{V = 0}$

$X_1 = 1 \qquad X_2 = 67$

at $V = 0 \to 0 = t^2 - t + 2$

no solution

distance

$= |X_2 - X_1| = (67 - 1)$

$= 66\ \text{m}$

x, v, a

rest $\rightarrow V=0$ (6)

$x=0$ $\quad$ $t=0$

A particle moves from origin with an initial velocity 6 m/s and moves along straight line according to the relation of $(v = 6 - k\,t^2)$ .Knowing that when t = 4 s, x = 13 m,

(a) when the velocity is zero.

(b) Determine the acceleration at this time.

(c) Determine the displacement and total distance from t =1 s to t = 6 s

t

$x=0 \rightarrow V=6 \rightarrow t=0$

$\boxed{x=13 \rightarrow t=4}$

$$\boxed{V=6-Kt^2}$$

$$\boxed{a=-K.(2t)}$$

$$x\Big|_0^x = 6t - K\frac{t^3}{3}\Big|_0^t$$

$$x-0 = 6t - K\frac{t^3}{3} - 0$$

$$\boxed{x = 6t - K\cdot\frac{t^3}{3}}$$

$$13 = 6(4) - K\,\frac{(4)^3}{3}$$

$$\boxed{K=0.5}$$

$$x = 6t - 0.5\,\frac{t^3}{3}$$

$$V = 6 - 0.5t^2$$

$$a = -t$$

(a) $V=0 \rightarrow t=??$

$0 = 6-0.5t^2 \rightarrow \boxed{t=3.46}$

(b) $t=3.46$

$a = -3.46\ m/s^2$

(c) $t=1$ $\quad$ $t=3.46$ $\quad$ $t=6$

$x_1 = 5.8$ $\quad$ $\boxed{V=0}$ $\quad$ $x_2 = 13.8$ $\quad$ $x_3 = 0$

$0 = 6-0.5t^2 \rightarrow t = 3.46$

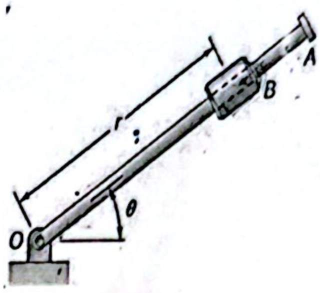
distance $= |0-13.8| + |13.8-5.8|$

$= 13.8 + 8 = 21.8$

disp $= 0-5.8 = -5.8\,m$

degree $\xrightarrow{\times \frac{\pi}{180}}$ rad

The rotation of the 0.9 m arm OA about O is defined by the relation $\theta = 0.15t^2$, where θ is expressed in radians and t in seconds. Collar B slides along the arm in such a way that its distance from O is $r = 0.9 - 0.12t^2$, where r is expressed in meters and t in seconds. After the arm OA has rotated through 30°, determine (*a*) the total velocity of the collar, (*b*) the total acceleration of the collar, (*c*) ~~the relative acceleration of the collar with respect to the arm.~~



Position $r = 0.9 - 0.12t^2$ $\qquad \theta = 0.15t^2$

Velocity $\dot{r} = -0.24t$ $\qquad \dot{\theta} = 0.3t$

acc $\ddot{r} = -0.24$ $\qquad \ddot{\theta} = 0.3$

Collar $\rightarrow r$

at $\theta = 30° \times \frac{\pi}{180} = \frac{1}{6}\pi$ rad $\rightarrow \frac{1}{6}\pi = 0.15t^2$

$\boxed{t = 1.86 \text{ sec}}$

$r = 0.48 \qquad \theta = 0.518$

$\dot{r} = -0.44 \qquad \dot{\theta} = 0.558$

$\ddot{r} = -0.24 \qquad \ddot{\theta} = 0.3$

$\rightarrow V_r = \dot{r} = -0.44 \text{ m/s}$ $\qquad \rightarrow V_\theta = r\dot{\theta} = 0.26$

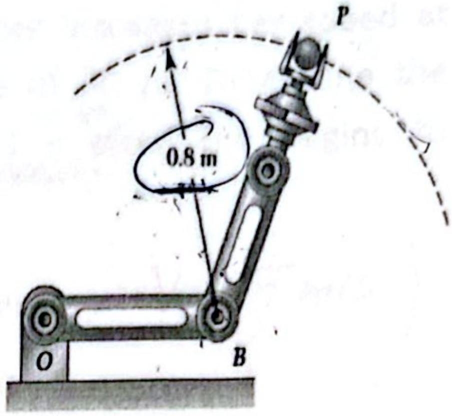
$\rightarrow a_r = \ddot{r} - r\dot{\theta}^2 = -0.389$ $\qquad \rightarrow a_\theta = 2\dot{r}\dot{\theta} + r\ddot{\theta} = -0.34$

(a) $V = \sqrt{V_r^2 + V_\theta^2} = 0.51 \text{ m/s}$

(b) $a = \sqrt{a_r^2 + a_\theta^2} = 0.516 \text{ m/s}^2$

(c) ~~$= -0.24 \text{ m/s}^2$~~

3) A robot arm moves so that P travels in a circle about point B, which is not moving. Knowing that P starts from rest, and its speed increases at a constant rate of 10 mm/s², determine (a) the magnitude of the acceleration when t = 4 s, (b) the time for the magnitude of the acceleration to be 80 mm/s².



$\rho = 0.8\,m \times 10^3$, $V_0 = 0$, $a_t = 10\ mm/s^2$

$= 800\,mm$

(a) $a_{total} = ??$ at $t = 4\ sec$

$\rightarrow V_f = V_0 + (a_t)\,t$

$= 0 + (10)(4) = 40\ mm/s^2$

$\rightarrow a_n = \frac{V^2}{\rho} = \frac{40^2}{800} = 2\ mm/s^2$

$\rightarrow a = \sqrt{a_t^2 + a_n^2} = 10.19\ mm/s^2$

(b) $t = ??$ at $a = 80\ mm/s^2$

$\rightarrow a = \sqrt{a_t^2 + a_n^2}$

$80 = \sqrt{10^2 + a_n^2}$ $\therefore a_n = 79.37\ mm/s^2$

$\rightarrow a_n = \frac{V^2}{\rho}$, $79.37 = \frac{V^2}{800}$

$\therefore V = 251.9\ mm/s$

$\rightarrow V = V_0 + (a_t)\,t$

$251.9 = 0 + (10)\,t$ $\boxed{t = 25.1\ sec}$

4) An outdoor track is 125 m in diameter. A runner increases her speed at a constant rate from 4 to 7 m/s over a distance of 30 m. Determine the magnitude of the acceleration of the runner 2 s after she begins to increase her speed.

$\rho = \frac{125}{2} = 62.5\ m$

$\left(V_0 = 4\ m/s,\ V_f = 7\ m/s,\ S = 30\ m, \right)$

$a = ??$ at $t = 2$ sec

to get a_t :

$V_f^2 = V_0^2 + 2(a_t)S$

$7^2 = 4^2 + 2(a_t)(30)$

$\boxed{a_t = 0.55\ m/s^2}$

to get a_n :

$a_n = \frac{V^2}{\rho}$

$V = V_0 + (a_t)t$

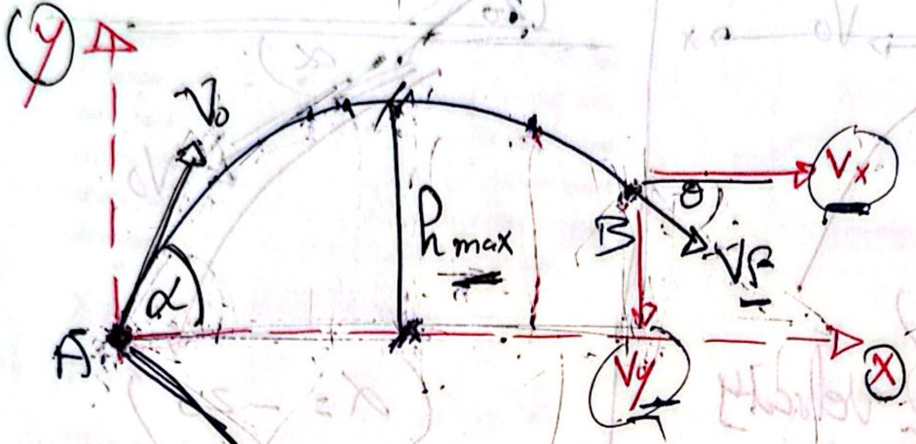
$= 4 + (0.55)(2) = 5.1\ m/s$

$a_n = \frac{5.1^2}{62.5} = \boxed{0.416\ m/s^2}$

$a = \sqrt{a_t^2 + a_n^2} = \boxed{0.68\ m/s^2}$

Projectiles

X, Y, V_0, α



α : (X-axis) vs (V_0) زاوية ميل

θ : (X-axis) vs (V_f) " "

h_{max} : max. height from (X-axis)

(X,Y) : احداثيات نقطة النهاية

initial horizontal velocity

$\boxed{\alpha = 0}$

Rules :

time →

$x = (V_0 \cos\alpha)\, t$

$y = (V_0 \sin\alpha)\, t - \frac{1}{2} g t^2$

$g \to 9.81\ m/s^2$

★★★ → $y = x \cdot \tan\alpha - \frac{1}{2} g \left(\frac{x^2}{V_0^2 \cdot \cos^2\alpha} \right)$ $\boxed{\text{Path. eq}}$

(x, y, V_0, α)

→ max. height : $h_{max} = \frac{V_0^2 \cdot \sin^2\alpha}{2g}$ ($g \to 9.81$)

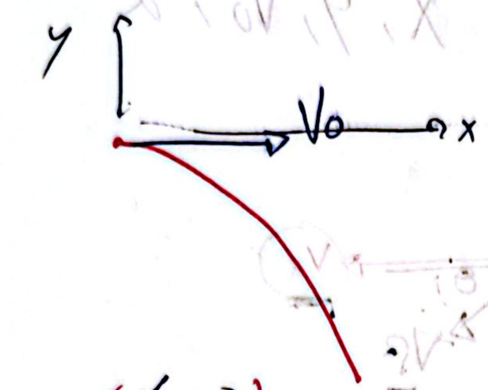
من X-axis

→ final Velocity : → $V_x = V_0 \cos\alpha$

→ $V_y = V_0 \sin\alpha - g t$

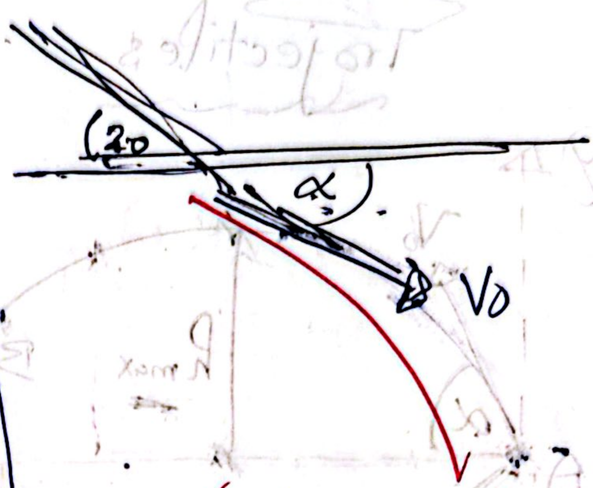
→ $V_f = \sqrt{V_x^2 + V_y^2}$

→ direction : → $\theta = \tan^{-1}(V_y / V_x)$

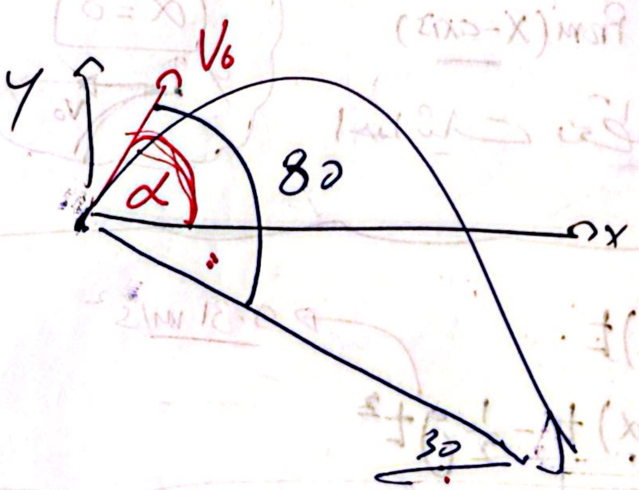


$(\alpha = 0)$

Horizontal Velocity

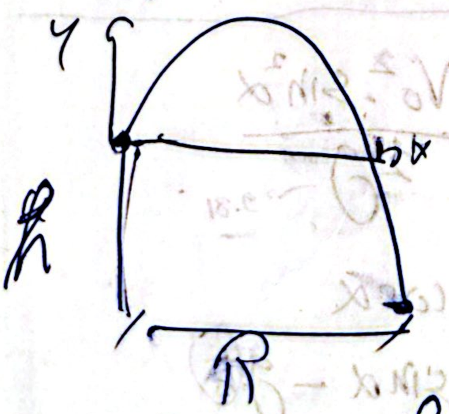


$(\alpha = -20)$

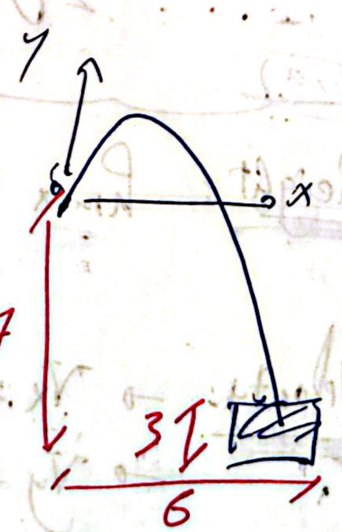


$\alpha = 80 - 30$

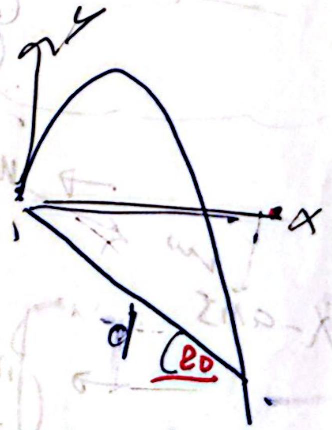
$(= 50)$



$x = R \quad y = -h$

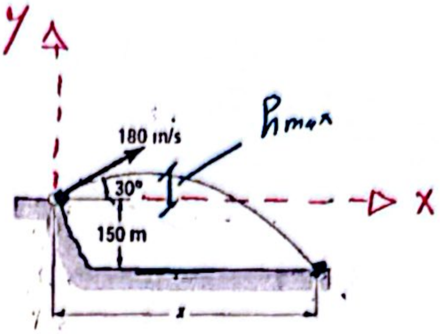


$x = 6 \quad y = -4$



$x = d\cos 20$

$y = -d\sin 20$



A projectile is fired from the edge of a 150-m cliff with an initial velocity of 180 m/s at an angle of 30° with the horizontal. Neglecting air resistance, find (a) the horizontal distance from the gun to the point where the projectile strikes the ground, (b) the greatest elevation above the ground reached by the projectile.

$X = ??$

$V_0 = 180\ m/s$

$\alpha = 30^\circ$

$y = -150\ m$

$\rightarrow y = x \cdot \tan\alpha - \frac{1}{2} g \frac{x^2}{V_0^2 \cos^2\alpha}$

$-150 = x \cdot \tan 30 - \frac{1}{2}(9.81) \frac{x^2}{180^2 \cos^2 30}$

$x = -239.7 \rightarrow$ (refused)

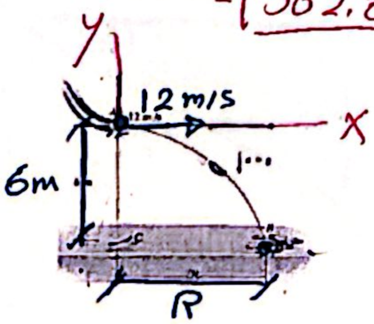
$\boxed{x = 3100m}$

$\rightarrow h_{max} = \frac{V_0^2 \cdot \sin^2\alpha}{2g} = \frac{180^2 \sin^2 30}{2(9.81)} = 412.8m$

$\rightarrow$ greatest elevation from the ground

$= 412.8 + 150$

$= \boxed{562.8m}$



A sack slides off the ramp, shown in Fig., with a horizontal velocity of 12 m/s. If the height of the ramp is 6 m from the floor, determine the time needed for the sack to strike the floor and the range R where sacks begin to pile up.

$V_0 = 12\ m/s$, $\alpha = 0$

$x = R$, $y = -6$, $t = ??$, $R = ??$

$\rightarrow y = x \cdot \tan\alpha - \frac{1}{2} g \frac{x^2}{V_0^2 \cos^2\alpha}$

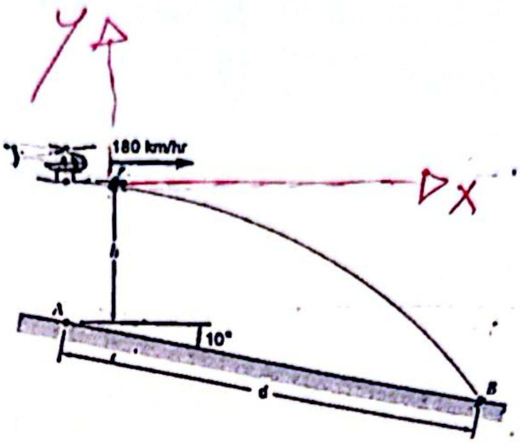
$-6 = R \cdot \tan 0 - \frac{1}{2}(9.81) \frac{R^2}{12^2 \cos^2 0}$

$\boxed{R = 13.27m}$

$\rightarrow x = (V_0 \cos\alpha) t$

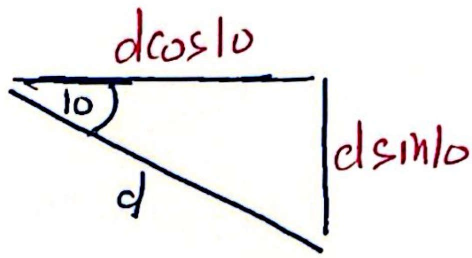
$13.27 = (12 \cos 0) t \longrightarrow \boxed{t = 1.1 sec}$

2) A helicopter is flying with a constant horizontal velocity of 180 km/h and is directly above Point A when a loose part begins to fall. The part lands 6.5 s later at Point B on an inclined surface. Determine (a) the distance d between Points A and B, (b) the initial height h.



$V_0 = 180 * \frac{5}{18} = 50$ m/s $\qquad \alpha = 0$

$t = 6.5$ sec



$X = d\cos 10$

$y = -h - d\sin 10$

$X = (V_0 \cos\alpha)t$

$d\cos 10 = (50\cos 0)(6.5) \longrightarrow d = \underline{330\,m}$

$y = (V_0 \sin\alpha)t - \frac{1}{2}gt^2$

$-h - d\sin 10 = (\cancel{180\sin 0)(6.5}) - \frac{1}{2}(9.81)(6.5)^2$

$d \rightarrow 330\,m$

$\longrightarrow h = \underline{149.9\,m}$

Newton's Second Law of motion

$$\Sigma F = m\,a$$

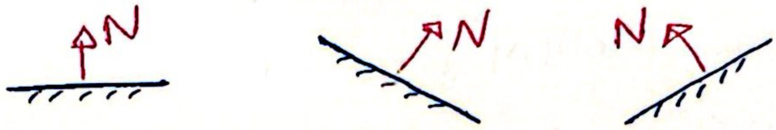
m → mass (Kg)

a → acceleration (m/s^2)

Types of forces :

① Weight $= m \cdot g$ (g = $9.81\ m/s^2$) ⟶ (رأسي لأسفل) ↓

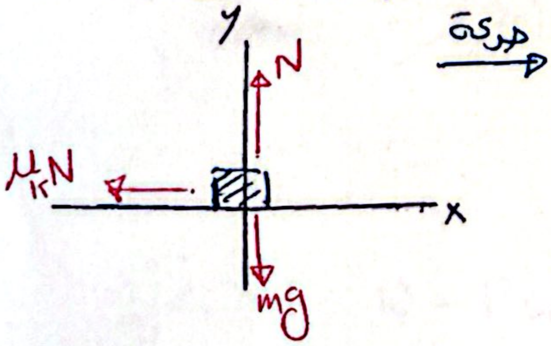
② Normal (N) ⟶ (عمودي على السطح لأعلى)



③ friction ($F_r = \mu_k N$) ⟶ (عكس اتجاه الحركة)

μ_k ⟶ coefficient of Kinetic friction

Horizontal Surface

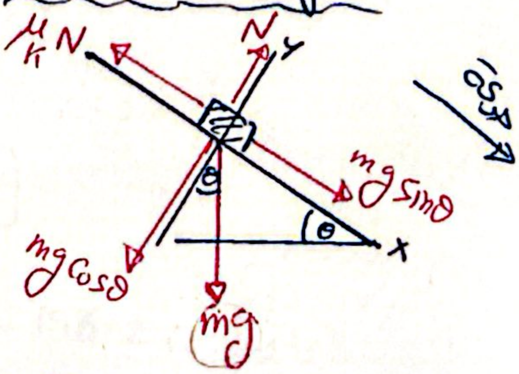


- $\Sigma F_y = 0, \quad N - mg = 0$
- $\Sigma F_x = ma_x, \quad -\mu_k N = m \cdot a_x$

$$-\mu_k (\cancel{m} g) = \cancel{m} a_x$$

$(\div m)$

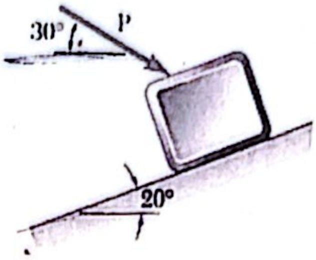
Inclined Surface



- $\Sigma F_y = 0, \quad N - mg\cos\theta = 0.$
- $\Sigma F_x = ma_x, \quad mg\sin\theta - \mu_k N = ma_x$

$$\cancel{m} g \sin\theta - \mu(\cancel{m} g \cos\theta) = \cancel{m} a$$

A 20-kg package is at rest on an incline when a force **P** is applied to it. Determine the magnitude of **P** if 10 s is required for the package to travel 5 m up the incline. The static and kinetic coefficients of friction between the package and the incline are both equal to 0.3.

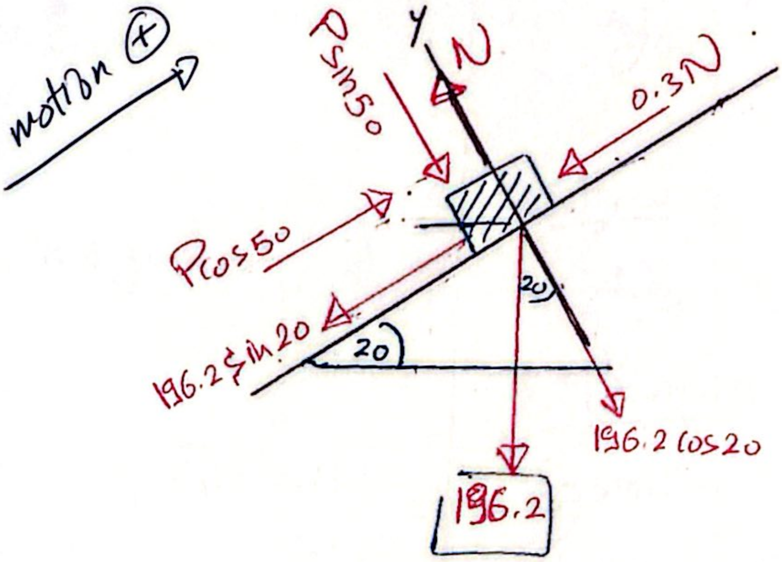


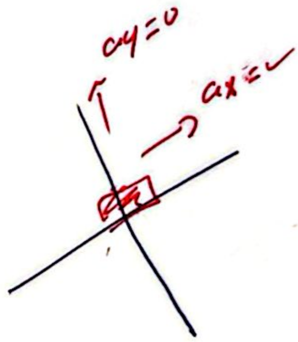
$m = 20\,kg$. $t = 10\,sec$, $S = 5\,m$, $V_0 = 0$

$\mu_s = \mu_K = 0.3$, $W = 20 \times 9.81 = 196.2\,N$

To get acceleration: $S = V_0 t + \frac{1}{2} a t^2$

$5 = 0 + \frac{1}{2}(a)(10)^2 \rightarrow a = 0.1\,m/s^2$





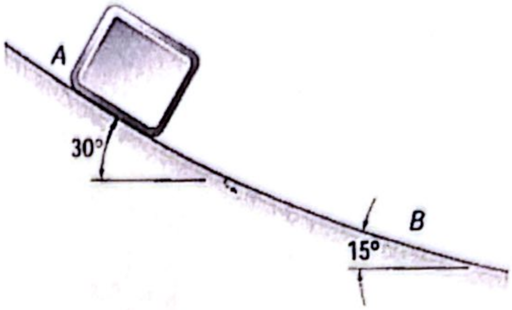
$\Sigma F_y = 0$: $N - P\sin 50 - 196.2\cos 20 = 0$

$\Sigma F_x = m a_x$: $-0.3N + P\cos 50 - 196.2\sin 20 = (20)(0.1)$

$P = 301.25\,N$

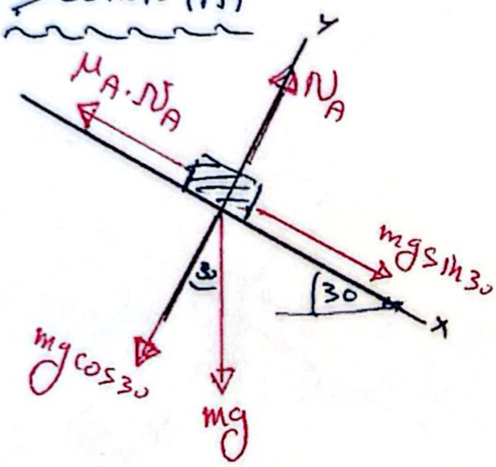
$N = 415.14\,N$

The acceleration of a package sliding at point A is 3 m/s². Assuming that the coefficient of kinetic friction is the same for each section, determine the acceleration of the package at point B.



$a_A = 3 m/s^2$, $\mu_A = \mu_B$

Section (A)



$\Sigma F_y = 0$:

$N_A = mg\cos 30$

$\Sigma F_x = m a_A$

$mg\sin 30 - \mu_A N_A = m(3)$

$\cancel{m}g\sin 30 - \mu_A(\cancel{m}g\cos 30) = \cancel{m}(3)$

$(\div m)$

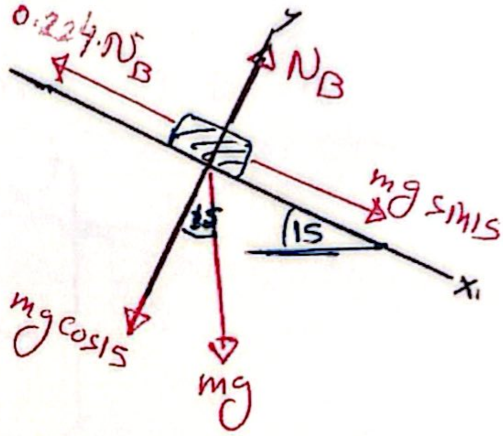
$g\sin 30 - \mu_A \cdot g\cos 30 = 3$

9.81 9.81

$\mu_A = 0.224$

$\mu_A = \mu_B = 0.224$

Section (B)



$\Sigma F_y = 0$

$N_B = mg\cos 15$

$\Sigma F_x = m a_B$

$mg\sin 15 - 0.224 N_B = m(a_B)$

$\cancel{m}g\sin 15 - 0.224(\cancel{m}g\cos 15) = \cancel{m}(a_B)$

$(\div m)$

$g\sin 15 - 0.224\, g\cos 15 = a_B$

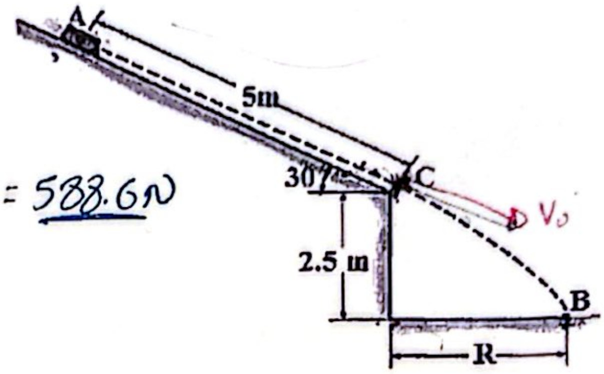
9.81 9.81

$\boxed{a_B = 0.416 m/s^2}$

*

↗ no friction

1) A 60-kg suitcase slides from rest 5 m down the smooth ramp. Determine the distance R where it strikes the ground at B. How long does it take to go from A to B?



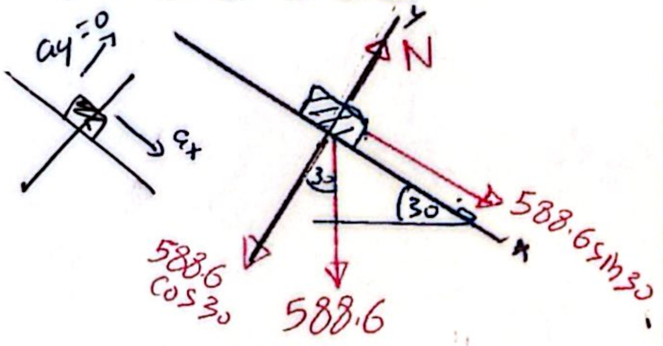
$m = 60\,kg$, $V_0 = 0$

$S = 5\,m$, $W = 60 \times 9.81 = 588.6\,N$

smooth $\rightarrow$ $(\mu N = 0)$

$R = ??$, $t_{A \to B} = ??$

From (A) to (C)



$\Sigma F_x = m a_x$

$588.6 \sin 30 = 60\,a$

$\boxed{a = 4.9\ m/s^2}$

- $V_c^2 = V_0^2 + 2 a S$

  $V_c^2 = 0 + 2(4.9)(5)$

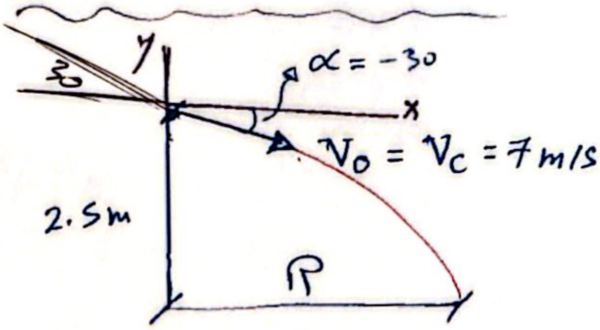
  $\boxed{V_c = 7\ m/s}$

- $V_c = V_0 + a\,t_{AC}$

  $7 = 0 + (4.9)\,t_{AC}$

  $\boxed{t_{AC} = 1.43\ sec}$

From (C) to (B)



$x = R$, $y = -2.5\,m$

- $y = x \cdot \tan\alpha - \frac{1}{2} g \frac{x^2}{V_0^2 \cos^2\alpha}$

  $-2.5 = R \cdot \tan(-30) - \frac{1}{2}(9.81) \frac{R^2}{7^2 \cdot \cos^2(-30)}$

  $\boxed{R = 2.67\ m}$

- $x = (V_0 \cos\alpha)(t_{CB})$

  $2.67 = (7 \cos(-30))(t_{CB})$

  $\boxed{t_{CB} = 0.44\ sec}$

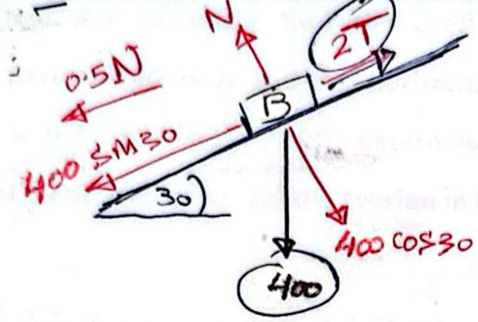
$t_{A \to B} = t_{AC} + t_{CB} = \boxed{1.87\ sec}$

$W = m\,g$

$400 = m\,g_{9.81}$

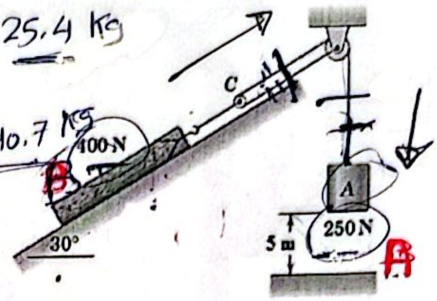
The 250-N concrete block A is released from rest in the position shown and pulls the 400-N log up the 30° ramp. If the coefficient of static and kinetic friction between the log and the ramp is 0.5, determine the velocity of the block as it hits the ground at B.

Block (B) :



$m_A = \frac{250}{9.81} = 25.4\ \text{kg}$

$m_B = \frac{400}{9.81} = 40.7\ \text{kg}$



$\Sigma F_y = 0 , \quad N = 400 \cos 30 = 346.4\ N.$

$\Sigma F_x = m_B\, a_B , \quad 2T - 0.5(346.4) - 400 \sin 30 = 40.7\, a_B$

$2T - 40.7\, a_B = 373.2 \longrightarrow eq\ ①$

Block (A) :

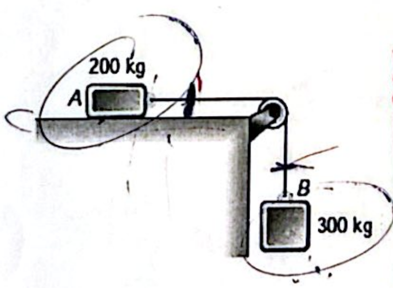
$2a_B = a_A$

$\Sigma F_y = m_A\, a_A$

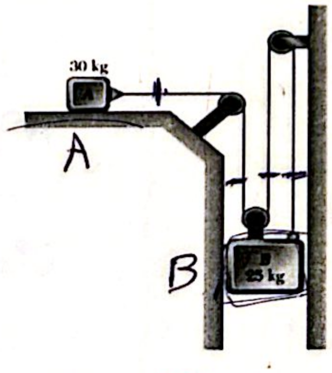
$250 - T = 25.4\, a_A$

$-T - 25.4\, \underset{(2a_B)}{a_A} = -250 \longrightarrow eq\ ②$

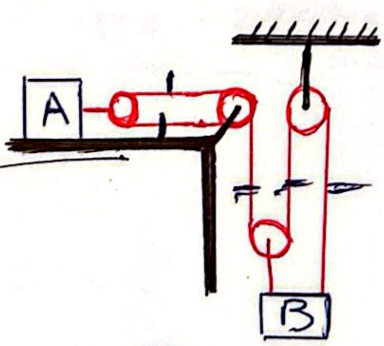
from eq ①, ② $\quad T = 204.7\,N \quad a_B = 0.89\ m/s^2 \quad a_A = 1.78\ m/s^2$



$a_A = a_B$



$a_A = 3a_B$

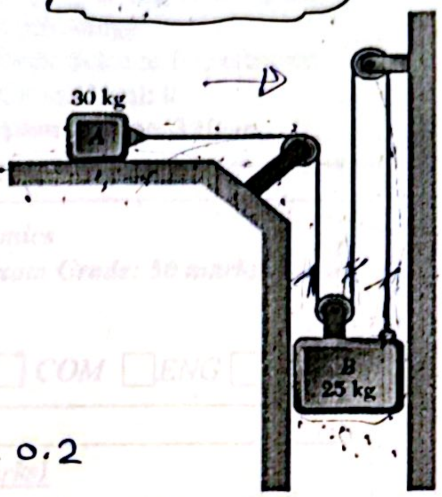


$2a_A = 3a_B$

(5)

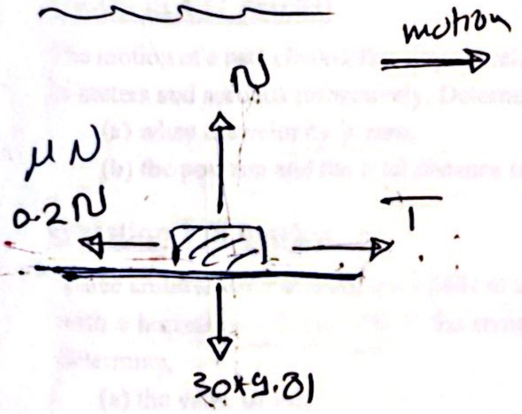
$\{a_A = 3a_B\}$

5) The two blocks shown are originally at rest. Neglecting the masses of the pulleys and the effect of friction in the pulleys and assuming that the coefficients of friction between block A and the horizontal surface are $\mu_s = 0.25$ and $\mu_k = 0.20$, determine (a) the acceleration of each block, (b) the tension in the cable.



$m_A = 30\,Kg$, $m_B = 25\,Kg$, $\mu_K = 0.2$

Block (A)



$\Sigma F_y = 0.$ $N - 30*9.81 = 0$

$N = 294.3\,N$

$\Sigma F_x = m a_A$

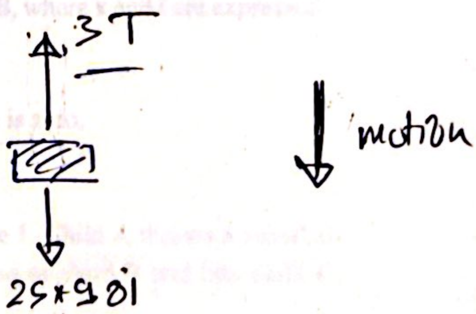
$T - 0.2\underbrace{N}_{294.3} = 30 a_A$

$T - 30\cancel{a_A} = 58.86$

$(3a_B)$

$\{T - 90a_B = 58.86\}$

Block (B)



$\Sigma F_y = m a_B$

$25*9.81 - 3T = 25a_B$

$\{-3T - 25a_B = -245.25\}$

$T = 80\,N$

$a_B = 0.23\,m/s^2$

$a_A = 3a_B = 0.69\,m/s^2$

BADR UNIVERSITY IN CAIRO
جامعة بدر بالقاهرة

Academic year: 2023/20234
Semester: 2nd semester
Exam: Final Term Exam
Code: BAS 032
Date: 4/6/2024

Faculty of Engineering and Technology
Basic Science Department
Course level: 0
Allowed Time: 3 Hour

Course Title: Mechanics (2) Dynamics

Number of Exam Questions: 5 ***Max Exam Grade: 50 marks***

Course Offered By

☒ *FRE* ☐ *STR* ☐ *ART* ☐ *ENR* ☐ *MET* ☐ *CMP* ☐ *COM* ☐ *ENG* ☐ *HUM*

Exam Instructions (plagiarized answers will receive zero marks)

1. Type of Exam: ☒ *Closed-Book* ☐ *Open-Book* ☐ *Charts and Formula Sheets Only*

2. Write your answers in: ☒ *Same Booklet* ☐ *Separate Booklet* ☐ *Drawing Sheet*

3. Attempt ALL questions

Question 1 (7 marks)

The motion of a particle is defined by the relation $x = t^3 - 9t^2 + 24t - 8$, where x and t are expressed in meters and seconds respectively. Determine:

(a) when the velocity is zero,

(b) the position and the total distance traveled when the acceleration is zero.

Question 2 (8 marks)

Three children are throwing snowballs at each other as shown in **Figure 1**. Child **A** throws a snowball with a horizontal velocity $V0$. If the snowball just passes over the head of child **B** and hits child **C**, determine,

(a) the value of $V0$,

(b) the horizontal distance **d**,

(c) the snowball velocity just before hits child **C**.

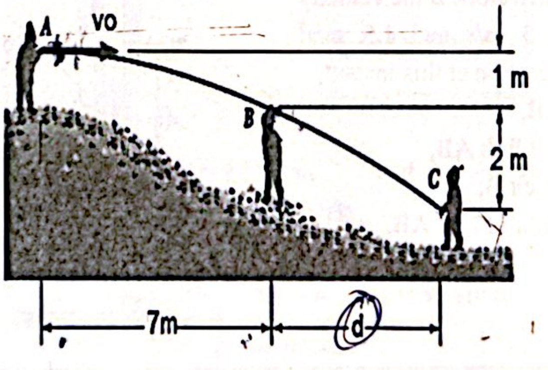


Figure 1

Question 3 (10 marks)

A 60-kg block slides from rest 7 m down the ramp as shown in Figure 2, the coefficient of kinetic friction between the suitcase and the ramp is $\mu_k = 0.2$. Determine:

(a) the block's velocity just before it reaches point C,
(b) the distance R where the block strikes the ground at B.

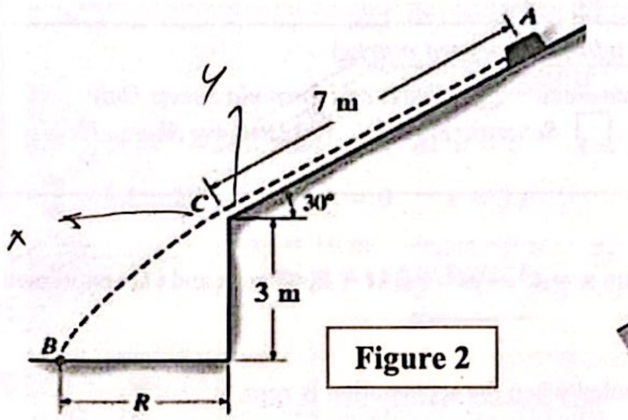


Figure 2

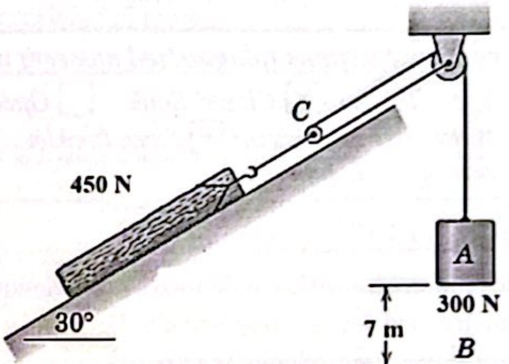


Figure 3

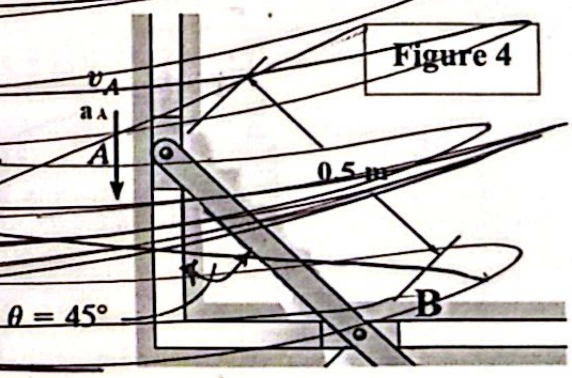
Question 4 (10 marks)

The 300-N concrete block A in Figure 3 is released from rest in the position shown and pulls the 450-N log up the 30° ramp. If the coefficient of static and kinetic friction between the log and the ramp is 0.2 and 0.3 respectively, determine the velocity and acceleration of block A as it hits the ground at B.

Question5 (15 marks)

The link shown in Figure 4 is guided by two blocks at A and B, which move in fixed slots. If the velocity and acceleration of A are 3 m/s and 1.5 m/s² downward respectively. Determine at this instant,

(a) the velocity of block B,
(b) the angular velocity of link AB,
(c) the acceleration of block B,
(d) the angular acceleration of link AB.



End the questions

With best wishes *Dr. Ahmed El-Desouky*

BUC BADR UNIVERSITY IN CAIRO جامعة بدر بالقاهرة	**Academic year: 2023/20234** **Semester: 2nd semester** **Exam: Final Term Exam** **Code: BAS 032** **Date: 4/6/2024**	**Faculty of Engineering and Technology** **Basic Science Department** **Course level: 0** **Allowed Time: 3 Hour**

Model Answer

Question 1 (7 marks)

The motion of a particle is defined by the relation $x = t^3 - 9t^2 + 24t - 8$, where x and t are expressed in meters and seconds respectively. Determine:

(a) when the velocity is zero,

(b) the position and the total distance traveled when the acceleration is zero.

Solution

a) $v = \frac{dx}{dt} = 3t^2 - 18\,t + 24$ $\quad$ $at\ v = 0$ $\quad$ $t = 2\,s\ and\ t = 4\,s$

b) $a = \frac{dv}{dt} = 6\,t - 18$ $\quad$ $at\ a = 0$ $\quad$ $t = 3\,s$

$$x_3 = 10\,m \quad x_0 = -8\,m \quad x_2 = 12\,m$$

$$T.D.T = |x_2 - x_0| + |x_3 - x_2| = 22m$$

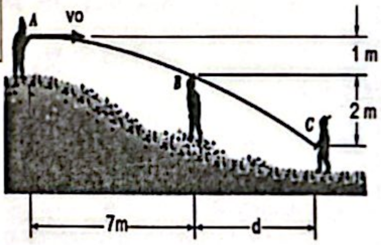
Question 2 (8 marks)

Three children are throwing snowballs at each other as shown in **Figure 1**. Child **A** throws a snowball with a horizontal velocity $\boldsymbol{V0}$. If the snowball just passes over the head of child **B** and hits child **C**, determine,

(a) the value of $\boldsymbol{V0}$, (b) the horizontal distance **d**,

(c) the snowball velocity just before hits child **C**.

Figure 1



Solution

a) From A to B $\quad$ y=-1m, x=7m, $\alpha = 0$

$$y = x\tan\alpha - \frac{1}{2} * 9.81\frac{x^2}{v_0^2\,(\cos\alpha)^2} \qquad \therefore v_0 = 15.5\,m/s$$

b) From A to c $\quad$ y=-3m, v_0=15.5m/s , $\alpha = 0$

$$y = x\tan\alpha - \frac{1}{2} * 9.81\frac{x^2}{v_0^2\,(\cos\alpha)^2} \qquad \therefore x = 12.12\,m \qquad \therefore d = x - 7 = 5.12\,m$$

C) at point C

$$v_x = v_0\cos\alpha = 15.5\frac{m}{s} \qquad t = \frac{x}{v_0\cos\alpha} = 0.78\,s \qquad v_y = v_0\sin\alpha - 9.81t = 7.67\ m/s$$

$$v = \sqrt{v_x^2 + v_y^2} = 17.3\,m/s$$

BAD'R UNIVERSITY IN CAIRO
جامعة بـــدر بالقاهرة

Academic year: 2024/2025
Semester: Spring
Exam: Midterm Exam
Code: BAS 032
Date: 24 /3/2025

Faculty of Engineering and Technology
Basic Science Department
Course level: 0
Allowed Time: 1 hour

Course Title: Mechanics (2)

Number of Exam Questions: 2 *Max Exam Grade: 15 marks*

Course Offered By

☒ *FRE* ☐ *STR* ☐ *ART* ☐ *ENR* ☐ *MET* ☐ *CMP* ☐ *COM* ☐ *ENG* ☐ *HUM*

Exam Instructions (plagiarized answers will receive zero marks)

1. Type of Exam: ☒ *Closed-Book* ☐ *Open-Book* ☐ *Charts and Formula Sheets Only*

2. Write your answers in: ☒ *Same Booklet* ☐ *Separate Booklet* ☐ *Drawing Sheet*

3. Attempt ALL questions

Question-1 (6 marks) (CLO 1)

The driver of a car decreases her speed at a constant rate from 20 to 10 m/s over a distance of 230 m along a curve of 150 m radius.

Determine :-

(a) The magnitude of the acceleration of the car 6s after using the brake.

(b) The time at which the total acceleration of the car is 0.8 m/s^2.

Question-2 (9 marks) (CLO 2)

The 5 Kg box shown in **Figure 1** falls off the conveyor belt at 1.5 m/s from A. If the coefficient of static and kinetic friction along *AB surface* is $\mu_s = 0.4$ and $\mu_k = 0.25$,

Determine,

(a) The acceleration of the box along the surface AB

(b) The velocity of the box at B.

(c) The distance x when the box falls into the cart at C.

(d) The velocity of the box just before it strikes the cart at C.

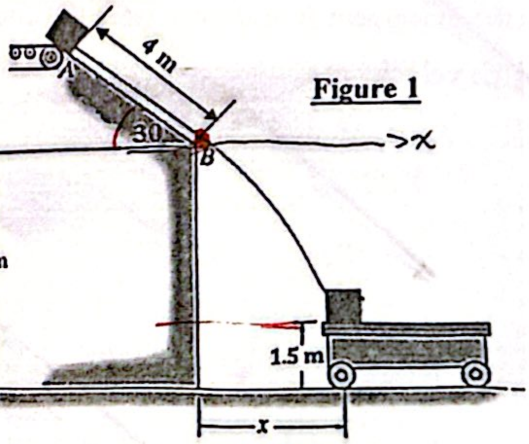


Figure 1

$v_f^2 = v_0^2 + 2ax$

$y = x\tan\theta - \frac{1}{2}g\left(\frac{x^2}{v_0^2\cos^2\theta}\right)$

With best wishes *Dr. Ehab Magdy*

BUC
BADR UNIVERSITY IN CAIRO
جامعة بدر بالقاهرة

Academic year: 2024/2025
Semester: 2nd semester
Exam: Final Term Exam
Code: BAS 032
Date: 28/5/2025

Faculty of Engineering and Technology
Basic Science Department
Course level: 0
Allowed Time: 3 Hour

Course Title: Mechanics (2) Dynamics
Number of Exam Questions: 5 *Max Exam Grade: 50 marks*

Course Offered By

☒ *FRE* ☐ *STR* ☐ *ART* ☐ *ENR* ☐ *MET* ☐ *CMP* ☐ *COM* ☐ *ENG* ☐ *HUM*

Exam Instructions (plagiarized answers will receive zero marks)

1. Type of Exam: ☒ *Closed-Book* ☐ *Open-Book* ☐ *Charts and Formula Sheets Only*
2. Write your answers in: ☒ *Same Booklet* ☐ *Separate Booklet* ☐ *Drawing Sheet*
3. Attempt ALL questions

Question 1 (10 marks) (CLO 1)

The motion of a particle is defined by the relation $x = t^3 - 9t^2 + 24t - 8$, where x and t are expressed in meters and seconds respectively. **Determine:**

(a) when the velocity is zero,

(b) the **displacement** and the total distance traveled when the **acceleration is zero.**

Question 2 (10 marks) (CLO 3)

Small balles shown in **Fig. 1** having a mass of **0.005 Kg** are dropped from rest at A through the smooth glass tube and accumulate in the can at C. **Determine**:

(a) the velocity of the ball at the end of the tube at B.

(b) the placement R of the can from the end of the tube

(c) the velocity at C which the marbles fall into the can.

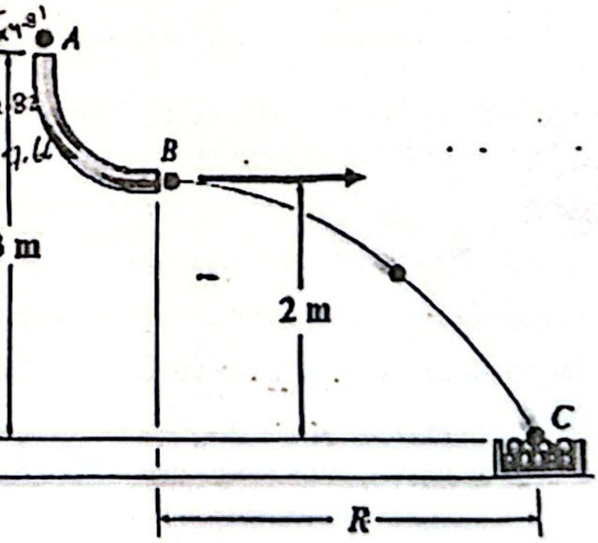


Figure 1

BUC
BADR UNIVERSITY IN CAIRO
جامعة بدر بالقاهرة

Academic year: 2024/2025
Semester: 2^{nd} semester
Exam: Final Term Exam
Code: BAS 032
Date: 28/5/2025

Faculty of Engineering and Technology
Basic Science Department
Course level: 0
Allowed Time: 3 Hour

Question 3 (10 marks) (CLO 2)

The two blocks shown in Fig. 2 are originally at rest. Neglecting the masses of the pulleys and the effect of friction in the pulleys and assuming that the coefficients of friction between block A and the horizontal surface are $\mu_s = 0.25$ and $\mu_k = 0.20$, **Determine:**

(a) the acceleration of each block, (b) the tension in the cable.

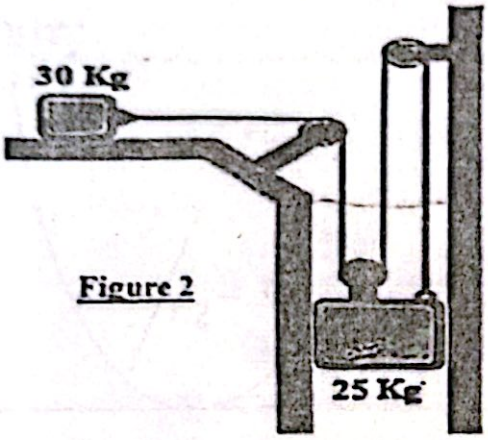


Figure 2

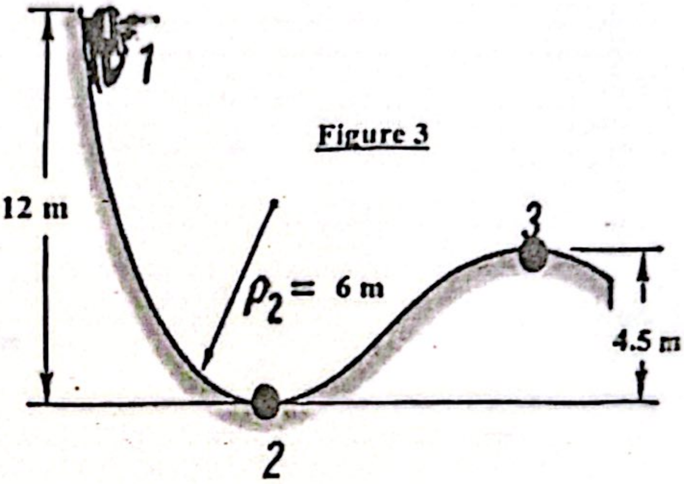


Figure 3

Question 4 (10 marks) (CLO 3)

A 1000 kg car starts from rest at point 1 and moves without friction down the roller coaster track shown in Fig. 3 **Determine**:

(a) the normal force (N) exerted by the track on the car at point 2,

(b) the minimum safe value of the **radius of curvature** at point 3.

Question 5 (10 marks) (CLO 4)

A spring shown in Fig. 4 is used to stop a 5-kg box which is moving down the inclined plane at a speed of 6 m/s. The coefficient of kinetic friction between the box and the plane is $\mu_k = 0.25$, and the spring constant is $k = 4$kN/m. **Determine** the deflection in the spring after the box hits it.

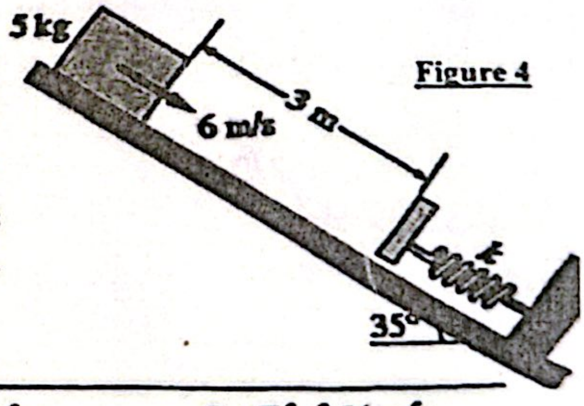


Figure 4

End the questions *With best wishes* *Dr. Ehab Magdy*

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BUC BADR UNIVERSITY IN CAIRO جامعة بدر بالقاهرة	Academic year: 2025/2026 Fall semester Exam: Final Term Exam Code: BAS 032 Date: 19/1/2026	Faculty of Engineering and Technology Basic Science Department Course level: 0 Allowed Time: 3 Hour

Course Title: Mechanics (2) Dynamics

Number of Exam Questions: 5 *Max Exam Grade: 50 marks*

Course Offered By

☒*FRE* ☐ *STR* ☐*ART* ☐*ENR* ☐ *MET* ☐ *CMP* ☐ *COM* ☐*ENG* ☐ *HUM*

Exam Instructions (plagiarized answers will receive zero marks)

1. Type of Exam: ☒ *Closed-Book* ☐ *Open-Book* ☐ *Charts and Formula Sheets Only*

2. Write your answers in: ☒ *Same Booklet* ☐ *Separate Booklet* ☐ *Drawing Sheet*

3. Attempt ALL questions

Question 1 (10 marks) (CLO 1)

The rotation of rod OA shown in **figure 1** about **O** is defined by the relation $\boldsymbol{\theta = (4\pi t^2 - 8\pi t)}$, where $\boldsymbol{\theta}$ and $\boldsymbol{t}$ are expressed in radians and seconds, respectively. Collar **B** slides along the rod so that its distance from O is $\mathbf{r = 10 + 6 \sin \pi t}$, where r and t are expressed in m and seconds, respectively. **When t = 1 s, determine**

(a) the velocity of the collar,

(b) the total acceleration of the collar.

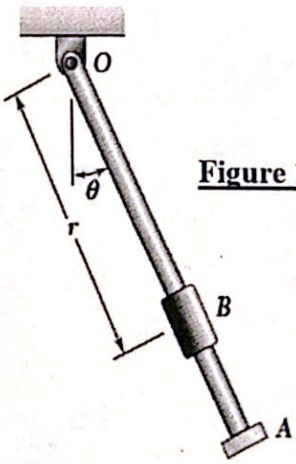


Figure 1

Question 2 (10 marks) (CLO 2)

A **20-kg** package shown in **figure 2** is at rest on an incline when a force **P** is applied to it. Determine the magnitude of **P** if **10 s** is required for the package to travel **5 m** up the incline. The static and kinetic coefficients of friction between the package and the incline are **0.4** and **0.3** respectively.

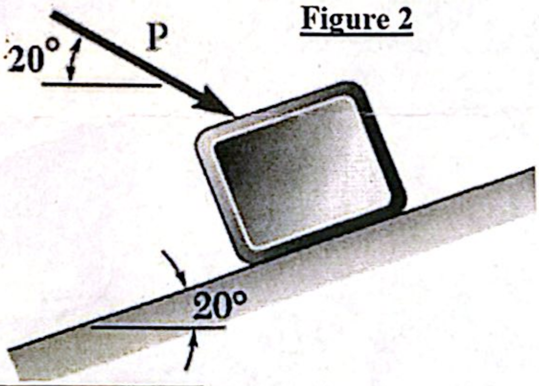
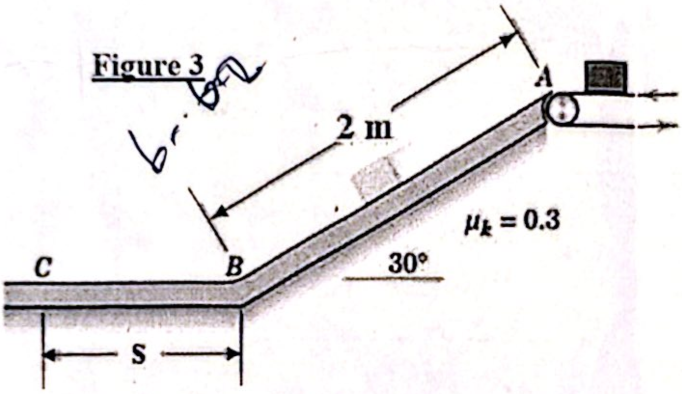


Figure 2

Question 3 (10 marks) (CLO 3)

A small package is deposited by the conveyor belt onto the 30^o ramp at A with a velocity of 0.8 m/s. The cofficent of kinetic friction for the package and the surface ABC is 0.3 **Determine** The distance S on the level surface BC at which the package comes to rest.

Figure 3



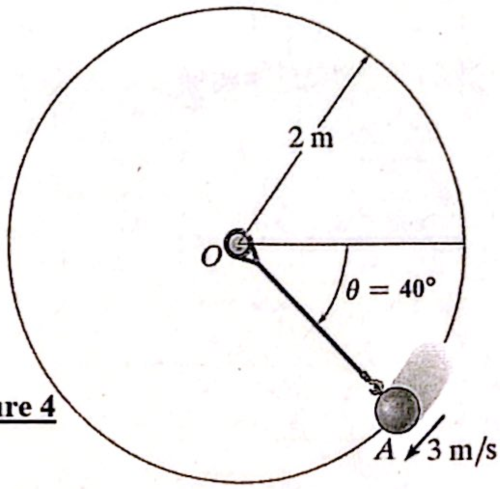


Figure 4

Question 4 (10 marks) (CLO 3)

If the **10-kg** ball shown in figure 4 has a velocity of **3 m/s** when it is at the position A, along the vertical circular path of radius **2 m**, **Determine,**

(a) the tension in the cord.

(b) the total acceleration of the ball when it reaches to point **A**.

Question 5 (10 marks) (CLO 4)

A **4-kg** collar shown in figure 5 is attached to a spring and slides without friction along a circular rod. The spring has an **un-deformed length of 0.2 m** and a constant **k = 400 N/m**. Knowing that the velocity of the collar is **2 m/s** when it is at **A, determine** the velocity of the collar at **B** if the circular rod,

(a) in a **vertical plane,**

(b) in a **horizontal plane.**

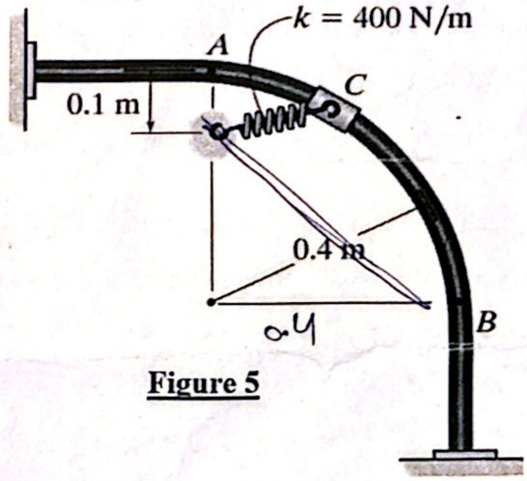


Figure 5

End the questions *With best wishes* *Dr. Ehab Magdy*